LAPRAK TEKNOLOGI IOT

TUGAS MODUL 4 & 5

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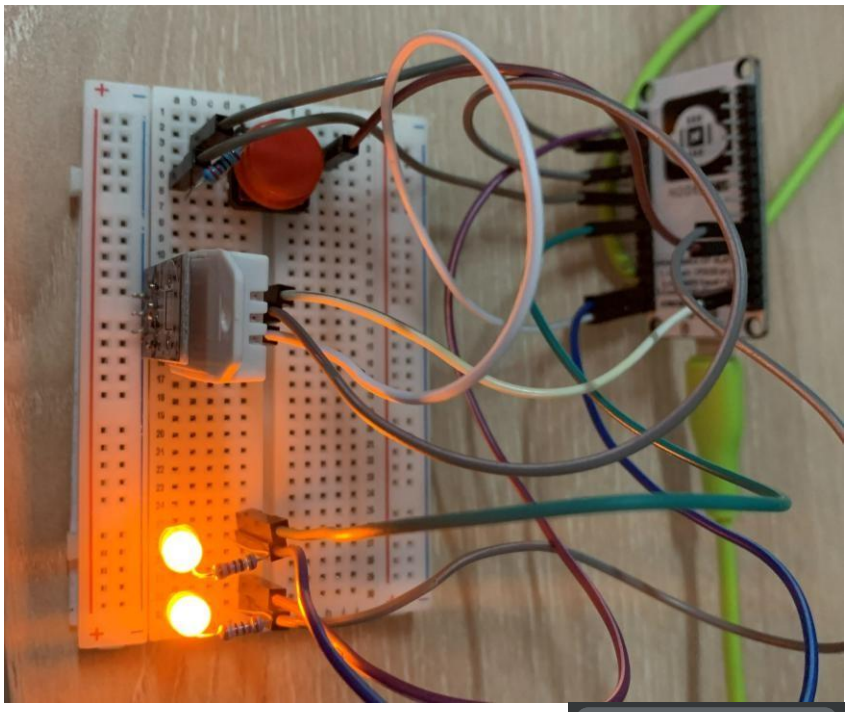
TEKNOLOGI REKAYASA INTERNET

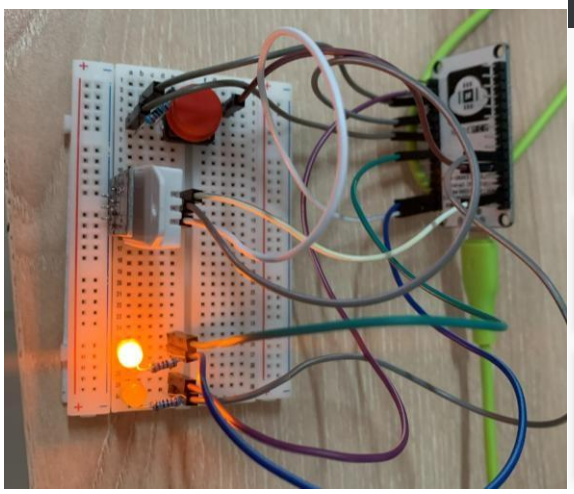
JURUSAN TEKNOLOGI INFORMASI

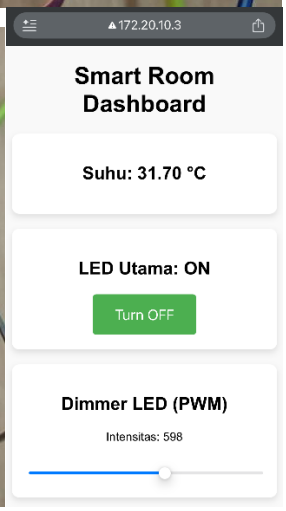
POLITEKNIK NEGERI LAMPUNG

2025/2026

4.7 Tugas: Sistem Kendali Intensitas Cahaya (PWM Slider) via *WebSockets*







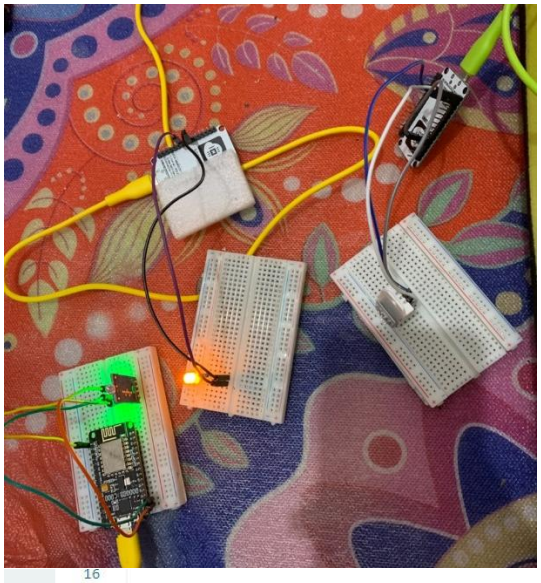
5.7 Tugas: Sistem Telemetri & Otomasi Terdistribusi Berbasis painlessMesh

```
11  DHT dht(DHTPIN, DHTTYPE);
12
13  Scheduler userScheduler;
14  painlessMesh mesh;
15
```

Output Serial Monitor ×

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')

```
[KIRIM Node 1] {"tipe":"suhu_node","suhu":33,"kelembapan":67.9}
[KIRIM Node 1] {"tipe":"suhu_node","suhu":33,"kelembapan":67.9}
[KIRIM Node 1] {"tipe":"suhu_node","suhu":33,"kelembapan":67.8}
[KIRIM Node 1] {"tipe":"suhu_node","suhu":33,"kelembapan":67.8}
[KIRIM Node 1] {"tipe":"suhu_node","suhu":33,"kelembapan":67.9}
[KIRIM Node 1] {"tipe":"suhu_node","suhu":33,"kelembapan":67.8}
[KIRIM Node 1] {"tipe":"suhu_node","suhu":33,"kelembapan":67.8}
[KIRIM Node 1] {"tipe":"suhu_node","suhu":33,"kelembapan":67.9}
[KIRIM Node 1] {"tipe":"suhu_node","suhu":33,"kelembapan":67.8}
[KIRIM Node 1] {"tipe":"suhu_node","suhu":33,"kelembapan":67.8}
[KIRIM Node 1] {"tipe":"suhu_node","suhu":33,"kelembapan":67.7}
[KIRIM Node 1] {"tipe":"suhu_node","suhu":33,"kelembapan":67.9}
```



```
16
17  void evaluasiRuleEngine() {
18    // RULE ENGINE DIREVISI:
19    // Nyala jika Suhu > 31.0 ATAU Cahaya > 700 (Angka gelap di rangkaian Anda)
20    if (suhuTerakhir > 31.0 || adcTerakhir > 700) {
21      digitalWrite(ledPin, HIGH); // Nyalakan LED
22      Serial.println("-> STATUS: LED NYALA (ON) [Kondisi Gelap / Panas]");
23    } else {
24      digitalWrite(ledPin, LOW);  // Matikan LED
25      Serial.println("-> STATUS: LED MATI (OFF) [Kondisi Normal / Terang]");
26    }
27  }
28
29  void receivedCallback(uint32_t from, String &msg) {
30    StaticJsonDocument<200> doc;
31    DeserializationError error = deserializeJson(doc, msg);
```

Output Serial Monitor ×

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')

```
[TERIMA] Suhu: 32.90 °C | -> STATUS: LED NYALA (ON) [Kondisi Gelap / Panas]
[TERIMA] Cahaya (ADC): 526 | -> STATUS: LED NYALA (ON) [Kondisi Gelap / Panas]
[TERIMA] Suhu: 32.90 °C | -> STATUS: LED NYALA (ON) [Kondisi Gelap / Panas]
[TERIMA] Cahaya (ADC): 524 | -> STATUS: LED NYALA (ON) [Kondisi Gelap / Panas]
[TERIMA] Suhu: 33.00 °C | -> STATUS: LED NYALA (ON) [Kondisi Gelap / Panas]
[TERIMA] Cahaya (ADC): 522 | -> STATUS: LED NYALA (ON) [Kondisi Gelap / Panas]
[TERIMA] Suhu: 33.00 °C | -> STATUS: LED NYALA (ON) [Kondisi Gelap / Panas]
[TERIMA] Cahaya (ADC): 508 | -> STATUS: LED NYALA (ON) [Kondisi Gelap / Panas]
```

Building sketch

Type here to search

```
10  Scheduler userScheduler;
11  painlessMesh mesh;
12
13  void sendMessage();
14  Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
15
16  void sendMessage() {
17    int adcValue = analogRead(ldrPin);
18
19    // Membuat format JSON sesuai instruksi Tugas 5.7
20    StaticJsonDocument<200> doc;
21    doc["tipe"] = "cahaya_node";
```

Output Serial Monitor ×

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')

```
[KIRIM Node 2] {"tipe":"cahaya_node","adc":538}
[KIRIM Node 2] {"tipe":"cahaya_node","adc":557}
[KIRIM Node 2] {"tipe":"cahaya_node","adc":528}
[KIRIM Node 2] {"tipe":"cahaya_node","adc":526}
[KIRIM Node 2] {"tipe":"cahaya_node","adc":524}
[KIRIM Node 2] {"tipe":"cahaya_node","adc":522}
[KIRIM Node 2] {"tipe":"cahaya_node","adc":508}
[KIRIM Node 2] {"tipe":"cahaya_node","adc":514}
[KIRIM Node 2] {"tipe":"cahaya_node","adc":630}
[KIRIM Node 2] {"tipe":"cahaya_node","adc":516}
```